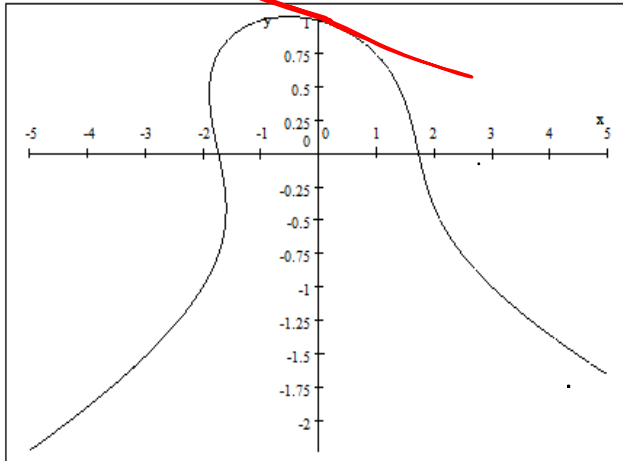


Unit 3.5 Implicit differentiation

Note Title

2012/04/12

Eg! $x^2 + xy + 3y^2 = 3$ — An unusual function



Finding the derivative:

$$x^2 + xy + 3y^2 = 3$$

$$\Rightarrow \frac{d}{dx} (x^2 + xy + 3y^2) = \frac{d}{dx} 3$$

$$\Rightarrow 2x + \underbrace{y + x \frac{dy}{dx}} + 6y \cdot \frac{dy}{dx} = 0$$

$$\Rightarrow \frac{dy}{dx} (x + 6y) = -2x - y$$

$$\Rightarrow \frac{dy}{dx} = \frac{-2x - y}{x + 6y}$$

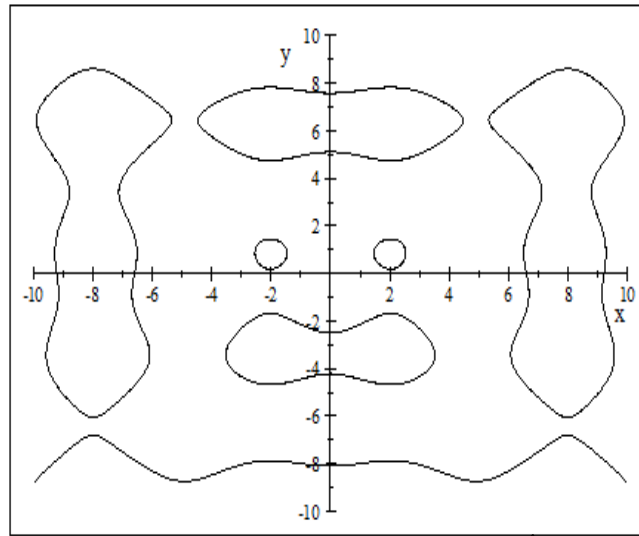
At $(0, 1)$ which is a point on the graph:

$$\frac{dy}{dx} = \frac{-1}{6}$$

Tangent: $y - 1 = -\frac{1}{6}(x - 0)$

$$\Rightarrow y = -\frac{1}{6}x + 1.$$

Eg 2 $x \sin x + y \cos y = 2$



$$\frac{d}{dx} (x \sin x + y \cos y) = \frac{d}{dx} 2$$

$$\Rightarrow \sin x + x \cos x + \frac{dy}{dx} \cos y + y (-\sin y \cdot \frac{dy}{dx}) = 0$$

$$\Rightarrow \frac{dy}{dx} (\cos y - y \sin y) = -\sin x - x \cos x$$

$$\Rightarrow \frac{dy}{dx} = \frac{-\sin x - x \cos x}{\cos y - y \sin y}$$

Eg 3. Finding the derivative of

$$x^3 + y^4 + x^5 y^7 = 8$$

$$\Rightarrow 3x^2 + 4y^3 \cdot \frac{dy}{dx} + 5x^4 \cdot y^7 + x^5 \cdot 7y^6 \frac{dy}{dx} = 0$$

$$\Rightarrow \frac{dy}{dx} = \frac{-3x^2 - 5x^4 y^7}{4y^3 + 7x^5 y^6}$$

Eg 4. The derivative of

$$\sin 2x = \cos 3y$$

$$\Rightarrow \cos 2x \cdot 2 = -\sin 3y \cdot 3 \frac{dy}{dx}$$

$$\Rightarrow \frac{dy}{dx} = \frac{-2 \cos 2x}{3 \sin 3y} \dots$$

Using implicit differentiation:

1. Finding the derivative of $y = \arcsin x$:
 $= \sin^{-1} x$

$$y = \arcsin x$$

$$\Rightarrow \sin y = x$$

$$y \in \left[-\frac{\pi}{2}, \frac{\pi}{2}\right]$$

$$\Rightarrow \cos y \cdot \frac{dy}{dx} = 1$$

$$\Rightarrow \frac{dy}{dx} = \frac{1}{\cos y} = \frac{1}{\sqrt{1 - \sin^2 y}}$$

$$= \frac{1}{\sqrt{1 - x^2}}$$

$$\Rightarrow \frac{d}{dx} \arcsin x = \frac{1}{\sqrt{1 - x^2}}$$

2. Finding the derivative of $y = \arctan x$:

$$y = \arctan x$$

$$\Rightarrow \tan y = x,$$

$$y \in \left(-\frac{\pi}{2}, \frac{\pi}{2}\right)$$

$$\Rightarrow \sec^2 y \cdot \frac{dy}{dx} = 1$$

$$\Rightarrow \frac{dy}{dx} = \frac{1}{\sec^2 y}$$

$$= \frac{1}{1 + \tan^2 y}$$

$$= \frac{1}{1 + x^2} \dots$$

Find the derivatives:

1. $y = \sin(x^2)$

$$\Rightarrow \frac{dy}{dx} = \cos(x^2) \cdot 2x$$

2. $y = \arcsin(x^2)$

$$\Rightarrow \frac{dy}{dx} = \frac{2x}{\sqrt{1-(x^2)^2}}$$

3. $y = e^{\arctan(x)}$

$$\Rightarrow \frac{dy}{dx} = e^{\arctan x} \cdot \frac{1}{1+x^2}$$

4. $y = \arctan(e^{2x})$

$$\Rightarrow \frac{dy}{dx} = \frac{1}{1+(e^{2x})^2} \times e^{2x} \cdot 2$$

5. $y = \arctan x \cdot \arcsin x$

$$\frac{dy}{dx} = \frac{1}{1+x^2} \cdot \arcsin x + \arctan x \cdot \frac{1}{\sqrt{1-x^2}}$$

In summary:

$$\frac{d}{dx} \sin^{-1}(f(x)) = \frac{f'(x)}{\sqrt{1-(f(x))^2}}$$

$$\frac{d}{dx} \tan^{-1}(f(x)) = \frac{f'(x)}{1+(f(x))^2}$$